CypherLink — Technical Architecture

CypherLink is designed as a collaboration-first underground creative ecosystem. The architecture prioritizes maintainability, low operational cost, explicit trust logic, and modular scalability.

Core Engineering Philosophy

- Modular monolith first
- Human-readable architecture
- Explicit state-driven systems
- Low infrastructure cost
- Replaceable providers and services
- No AI dependency for core workflows

Core Systems

- Role-based user system
- Structured and open collaboration projects
- Trust and collaboration memory graph
- Contextual reputation system
- Cypher/event coordination layer
- Lightweight project discussion threads
- Locality-biased discovery engine

Database Priorities

- Clear contributor relationships
- Project state machines
- Collaboration history
- Role occupancy tracking
- Trust event logging
- Soft reputation decay
- Event participation history

Security Considerations

- RBAC permission system

- Upload validation
- Rate limiting
- Private project controls
- Audit logging
- Safe event participation controls
- No public live location exposure

Infrastructure Strategy

- Frontend: Vercel or Cloudflare Pages
- Backend: modular API service
- Database: PostgreSQL
- Storage: compressed media-first approach
- Maps: OpenStreetMap/MapLibre
- No premature microservices